

1. The charge on a body is 8×10^{-12} C. It means that the body has :
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 - (b) gained 4×10^{10} electrons
 - (c) gained 2×10^8 electrons
 - (d) lost 5×10^7 electrons

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2. 10^9 electrons are transferred to a pith ball with charge 0.16 nC . Its charge now is :

- (a) Zero
- (b) $-3.2 \times 10^{-10} \text{ C}$
- (c) $-1.6 \times 10^{-9} \text{ C}$
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(a) Zero

2. Two identical small conducting balls B_1 and B_2 are given -7 pC and $+4$ pC charges respectively. They are brought in contact with a third identical ball B_3 and then separated. If the final charge on each ball is -2 pC, the initial charge on B_3 was

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(B) -3 pC

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(D) -15 pC

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1. A charge of $-1\ \mu\text{C}$ on a body represents **1**
- (A) loss of 6.25×10^{12} electrons by the body.
 - (B) gain of 6.25×10^{12} electrons by the body.
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1. A body acquires charge 8.0×10^{-12} C. The mass of the body :
- (A) increases by 4.5×10^{-7} kg
 - (B) decreases by 1.0×10^{-6} kg
 - (C) decreases by 4.55×10^{-23} kg
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Example 7: How much positive and negative charge is there in a cup of water (250 g)?. Given molecular mass of water is 18 g.

Example 8: A polythene piece rubbed with wool is found to have a negative charge of 3×10^{-7} C.

- (a) Estimate the number of electrons transferred (from which to which?)
- (b) Is there a transfer of mass from wool to polythene?